

MTH301 - Midsem 1

1. Define an ordered field $(\mathbb{K}, <)$. Let $\mathbb{F}$ be a field and $P \subseteq \mathbb{F}$. Suppose P is closed under addition and multiplication, and for each $x \in \mathbb{F} \setminus \{0\}$, either $x \in P$ or $-x \in P$. Show that $\mathbb{F}$ is an ordered field. Also, give an example of an ordered field which is non-Archimedean. Justify your answer. $(1 + 2 + 2 = 5)$
2. (a) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a field automorphism. Show that $f = 1_{\mathbb{R}}$ is the identity. (2)
 (b) Consider the open intervals $(1, 2)$ and $(-\infty, 2)$. Determine an explicit homeomorphism between these open intervals. Can the closed interval $[0, 1]$ and the open interval $(0, 1]$ be homeomorphic? Explain (Context: We are allowed to use the fact that intervals are connected) $(2 + 1 = 3)$

OR

Let $\mathbb{R}^2$ be the Euclidean plane and $p : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the map given by $p((x, y)) = xy$ for $(x, y) \in \mathbb{R}^2$. Show that p is continuous.

3. Define a metric space. If $A \neq \emptyset$ is a subset of a metric space (X, d) , and $d_A : X \rightarrow \mathbb{R}$ is given by

$$d_A(x) = \inf\{d(x, a) : a \in A\}, \quad \text{for } x \in X$$

Show that the closure $\overline{A} = \{x \in X : d_A(x) = 0\}$. $(2 + 3 = 5)$

OR

Show that the function $d_1(x, y) = \sqrt{|x - y|}$ for $x, y \in \mathbb{R}$ define a metric on $\mathbb{R}$. Consider the function $d_2(x, y) = |x - y| + \sqrt{|x - y|}$ for $x, y \in \mathbb{R}$. Is d_2 also a metric on $\mathbb{R}$? Explain. $(2 + 3 = 5)$

4. Let α and β be positive Dedekind cuts. Define the product $\alpha\beta$ of the positive Dedekind cuts α and β . If $1^* = \{r \in \mathbb{Q} : r < 1\}$ is the Dedekind cut corresponding to the rational number 1, then, show that $\alpha 1^* = \alpha$. Also describe a Dedekind cut δ in terms of α and verify that $\alpha\delta = 1^*$.